Eric Yu

Email: eyu36@jhu.edu
Website: https://e7yu.github.io/

Education

2025–2029 **Johns Hopkins University**, Baltimore, MD, USA
- **B.S.**, Molecular & Cellular Biology and Chemistry
- **GPA:** 4.00/4.00
- Dean's List: Fall 2025, Spring 2026

2023–2025 **Georgia Institute of Technology**, Atlanta, GA, USA
- Dual Enrollment in High School
- **GPA:** 4.00/4.00

My relevant coursework can be found at the end of my CV here.

Research Experience

2026– **Research Assistant**, Qiu Lab, Georgia Institute of Technology
- *Advisor:* Prof. Peng Qiu
- Developing a novel machine learning-based pipeline to analyze single-cell RNA sequencing data and characterize the associated phenotypes and biological features of cellular populations

2025– **Research Assistant**, Huang Lab, Johns Hopkins University
- *Advisor:* Prof. Xiongyi Huang
- Engineering proteins via directed evolution to access new-to-nature biocatalytic reactions, including azidation, CF_3 azidation, and skeletal editing reactions, enabling pharmaceutically important transformations
- Utilizing computational tools (PyMOL, AutoDock Vina) to analyze protein structure and substrate binding interactions
- Preparing substrate libraries through organic synthesis to evaluate the substrate scope of evolved enzymes

2024– **Research Assistant**, Mao Lab, Indiana University School of Medicine
- *Advisor:* Prof. Weiming Mao
- Designing a deep learning pipeline for automated segmentation and analysis of corneal endothelial cells and cross-linked actin networks with PyTorch
- Assisting with literature review and the analysis of medical data, including those from optical coherence tomography (OCT), fundus imaging, and cell scratch assays

Other Experience

2025– **Violinist**, Hopkins Symphony Orchestra

2023–2025 **Violinist**, Atlanta Symphony Youth Orchestra
- Selected by a competitive multi-round audition process
- Invited to perform at the 4th Annual Whitney E. Houston Foundation "Legacy of Love" Gala headlined by EGOT winner Jennifer Hudson

2022–2025 **Pianist & Violinist**, Georgia Philharmonic
- Performed Bernstein's *Symphonic Dances from West Side Story* as a piano soloist

2021–2023 **Violinist**, Emory Youth Symphony Orchestra

Awards & Honors

2025 **National Merit Scholar**, $2,500 Scholarship
- Selected from a pool of over 15,000 National Merit Finalists

2021–2025 **4-time National Medalist**, Science Olympiad National Tournament
- 8x Top 10 national tournament placements
- 4x State champion team and national qualifier
- Top 6, 9, and 14 national team placements as a leading member and captain

2024 **High Honors**, U.S. National Chemistry Olympiad (USNCO)
- Top 50 out of over 16,000 participants nationally (top ~0.3%) in the most prestigious national chemistry competition
- #1 in the state of Georgia on the Local Exam

2024 **Qualifier/Invitee**, American Invitational Mathematics Exam (AIME)
- Top scorer on the American Mathematics Competition (AMC)

2024 **Gold Award**, The President's Volunteer Service Award
- Awarded to those with 250+ volunteer hours in a 12 month period
- Previously won a silver award in 2022

2024 **Prize Winner**, Georgia Music Teacher's Association
- Award of Excellence in both the Instrumental Concerto and String competitions

2021–2023 **First Prize**, Sonatina & Sonata International Youth Piano Competition
- Prize winner of the XIV and XV editions
- 2x Carnegie Hall performance opportunities as winner

Volunteering

2025– **Event Supervisor**, Science Olympiad
- Test writer and grader for Maryland state and regional tournaments, Johns Hopkins invitational

2025– **Musicare**
- Music performance at local hospitals and nursing homes

2025– **MedShare & SHARE**
- Sorting medical supplies for distribution to developing nations

2025– **Mural Art**, Foundation for Hospital Art
- Painting murals to donate to hospitals and patients

2025– **Circle K International**
- Volunteering to help local communities in Baltimore

2022–2025 **Coach and Test Writer**, Science Olympiad
- Coached my middle school team in physics events, guiding them to win the state championship and a national silver medal
- Volunteered to write and grade tests for top invitational tournaments across the country

Skills

Programming & Software

- Programming Languages: Python, MATLAB
- Machine Learning/AI: PyTorch, scikit-learn
- Scientific & Technical Software: PyMOL, ImageJ, AutoDock Vina, ChemDraw, Microsoft Office 365
- Typesetting: LaTeX

Laboratory-related

- Organic Synthesis: Column chromatography, TLC, inert atmosphere (Schlenk line), liquid-liquid extractions, recrystallization, rotary evaporation
- Biomolecular Techniques: Molecular cloning and library construction (PCR, mutagenesis, gel electrophoresis), bacterial culture & transformation (electroporation), protein expression &

	purification (sonication, affinity chromatography, SDS-PAGE), enzymatic assays
– Analytical Techniques:	Chromatography (GC-MS, HPLC), spectroscopy (NMR, IR, UV-Vis, CD)

Publications

Detailed publication information can also be found at my Google Scholar G.

Journal Articles

J1. Dai, J., Rayana, N. P., Peng, M., Sugali, C. K., Harvey, D. H., Dhamodaran, K., **Yu, E.**, Dalloul, J. M., Liu, S. & Mao, W. Inhibition of pterygium cell fibrosis by the Rho kinase inhibitor. *Sci. Rep.* **14,** 30930 (2024). DOI.

Peer-reviewed Conference Proceedings & Published Abstracts

C1. Sugali, C. K., Dhamodaran, K., Dai, J., Harvey, D., Pushpala, N. M., **Yu, E.**, Huynh, K., Huang, A. Y. & Mao, W. Canonical Wnt signaling pathway inhibits glucocorticoid-induced ocular hypertension without compromising its anti-inflammatory function in the uveitis mouse model. *Invest. Ophthalmol. Vis. Sci.* **67,** 3010 (2026). LINK.
   – 2026 ARVO Annual Meeting in Denver, CO, May 3–7, 2026.

C2. Dai, J., Rayana, N. P., Peng, M., Sugali, C. K., Harvey, D. H., Dhamodaran, K., **Yu, E.**, Dalloul, J. M., Liu, S. & Mao, W. The Rho kinase inhibitor Y27632 inhibits unstimulated pterygium cell fibrosis. *Invest. Ophthalmol. Vis. Sci.* **66,** 6114 (2025). LINK.
   – 2025 ARVO Annual Meeting in Salt Lake City, UT, May 4–8, 2025.

Certifications

2026–2028 **Basic Life Support (BLS) Provider**, American Heart Association (AHA)
Credential ID: 271266216845

Relevant Coursework

Johns Hopkins University

- Physical Chemistry I (CHEM 301)
- Physical Chemistry II (CHEM 302)
- Intmd. Organic Chemistry Lab (CHEM 228)
- Genetics (BIOL 303)
- Biochemistry I (BPHY 315)
- Protein Eng. & Biochemistry Lab (BPHY 253)

- Physical Chem. & Instrum. Lab (CHEM 307)
- Biological Models & Simulations (BMED 242)
- Biological Physics (PHYS 310)
- Contemporary Physics Seminar (PHYS 203)

In progress:

- Qnt Analyt. Chem. & Spec. Lab (CHEM 246)
- Chem. of Inorganic Compounds (CHEM 449)
- Molecular Biology (BIOL 304)
- Intmd. Probability & Statistics (APPM 311)

Georgia Institute of Technology ***(dual enrolled in high school)***

- Cell & Molecular Biology (BIOS 3450)
- Organic Chemistry I (CHEM 2311)
- Organic Chemistry II (CHEM 2312)
- Synthesis Lab I (CHEM 2380)
- Intro to Modern Physics (PHYS 2213)
- Linear Algebra (MATH 1554)
- Multivariable Calculus (MATH 2551)
- Differential Equations (MATH 2552)
- Princ. & Appl. of Eng. Materials (MSE 2001)
- Intro to Computing (CS 1301)

Last updated: September 2, 2026